Technology stack and platforms

Programming languages and frameworks

Steppelight Technologies builds backend services mainly in Java 21 (Spring Boot 3), C# (.NET 8) and Python 3.12 (FastAPI, Django). Web front ends are written in TypeScript with React 18. Mobile applications are built natively in Kotlin and Swift, or with Flutter 3 when a single code base is preferred.

Data and messaging

The standard relational database is PostgreSQL 16. Oracle Database and Microsoft SQL Server are supported for customers with existing installations. For analytics the team uses ClickHouse and Apache Spark. Event streaming is based on Apache Kafka 3.7; caching on Redis 7.

Cloud and infrastructure

Workloads run on Kubernetes 1.30, provisioned with Terraform. Supported targets are AWS, Microsoft Azure and local data centres in Kazakhstan. Container images are built with Docker and stored in a private registry. Monitoring uses Prometheus and Grafana; logs are collected in an Elasticsearch-based stack.

Engineering tools

Source code is hosted in GitLab (self-managed), tasks are tracked in Jira, documentation is kept in Confluence. Secrets are stored in HashiCorp Vault. The company does not use technologies that are end-of-life in the vendor support lifecycle for new projects.